

Col	Row	MonitorName	Weight	Distance
1	246	410030013881011	0.215167	627.5258
1	246	530410006881011	0.253047	533.5881
1	246	530090009881011	0.286508	471.2713
2	246	410030013881011	0.213318	620.813
2	246	530410006881011	0.252881	523.688
2	246	530090009881011	0.287978	459.8632
3	246	410030013881011	0.211387	614.2613
3	246	530410006881011	0.252682	513.8765
3	246	530090009881011	0.289525	448.4836

The first two columns specify the *Column* and *Row* variables for each grid cell. In the example above, you will see that *Column* = 1 and *Row* = 246 are repeated three times, indicating that three different monitors were used to estimate air quality at this grid cell. The *MonitorName* column provides the monitor identifier. The *Weight* column specifies the weight used in the air quality calculation (e.g., *Voronoi Neighbor Averaging*). And the *Distance* column gives the distance (in kilometers) from the monitor location to the center of the grid cell.

Note that if an air quality grid was created using the *Closest Monitor* option (see Chapter 5 under the Monitor Data section), then only a single monitor is used for any given grid cell. As a result, the neighbor file will contain the same 5 fields, but the *Weight* column will contain a value of “1” all the way down. In addition, there will only be one entry for each grid cell.

Col	Row	MonitorName	Weight	Distance
1	246	410030013881011	1	627.5258
2	246	410030013881011	1	620.813
3	246	410030013881011	1	614.2613
4	246	410030013881011	1	607.8755
5	246	410030013881011	1	601.661
6	246	410671003881011	1	548.5376
7	246	410671003881011	1	541.038
8	246	410671003881011	1	533.702

9.10 PopSim

The Population Simulation (PopSim)³⁸ model was designed to estimate two outputs related to premature mortality attributed to the CAAA: life-years lost, and changes in life expectancy. The population simulation approach provides some advantages over the BenMAP-CE model in terms of simulation of the dynamic effects of mortality across a population through time, but also has several significant disadvantages relative to BenMAP-CE in terms of the spatial resolution of pollutant exposure estimates. As a result, the population simulation approach operates as a supplement to the BenMAP-based primary estimates for selected measures of the impact of reducing risks of premature mortality.

The core BenMAP-CE program estimates changes in adverse health effects based on changes in air quality for one specified analysis year, even though certain health benefits may occur after the analysis year. Conversely, the PopSim tool estimates the change in population mortality risk over a multi-year period, but it is not yet possible to estimate the economic value of these impacts in BenMAP-CE.

PopSim is designed to track the effect of alternative assumptions about the mortality effects of fine particulate matter (PM_{2.5}) air pollution in the U.S. population over time. The tool incorporates detailed life table data for historical years, by age, gender, and cause of death, obtained from the Census Bureau and the Centers for Disease Control (CDC). It also incorporates Census mortality and population projections for future years, again by age and gender, using the projected death and birth rates that underlie the Census Bureau's published population projections.

The PopSim model allows users to:

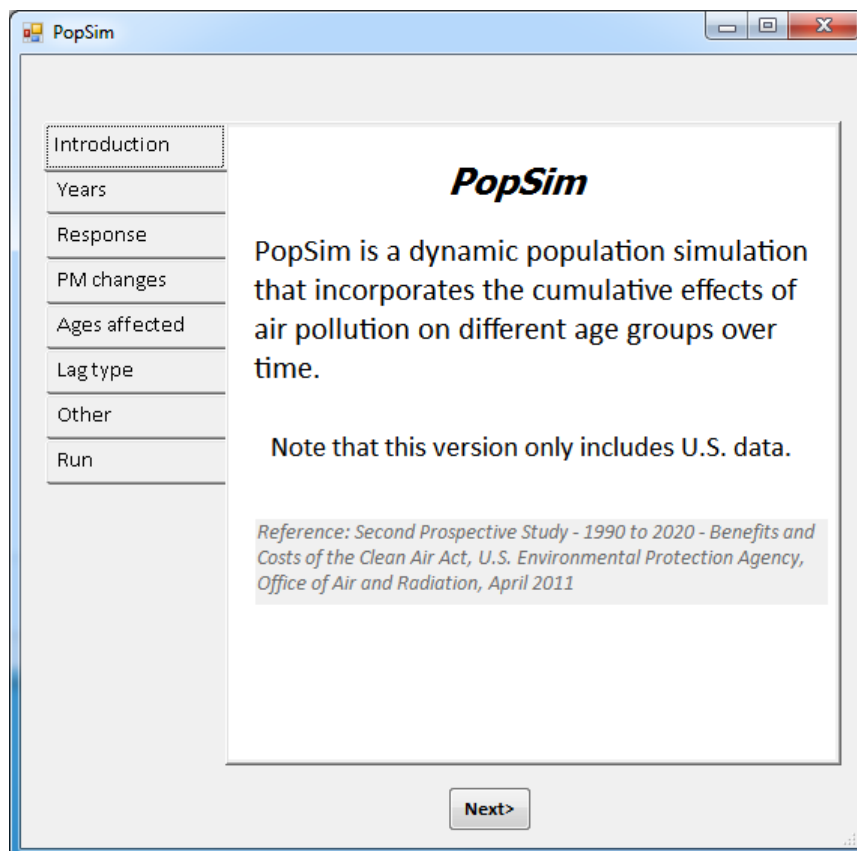
- Simulate population in the U.S. by single year cohorts of age and gender for years between 1980 and 2050 under alternative assumptions about the degree of hazard posed by air pollution relative to baseline historical and projected Census mortality rates;
- Estimate changes in life years relative to baseline Census mortality rates;
- Apply air pollution hazards differentially by cause of death; and
- Analyze the effect of alternative cessation lag structures on the timing of total mortality and on total life years in the U.S. population, based on differential application by cause of death or other specifications of cessation lag.

The dynamic life-table approach used in this model can theoretically provide improved estimates of the mortality impacts of air pollution in future years over the more common

³⁸ The PopSim tool in BenMAP-CE is based upon an Access-based model originally developed for EPA by Industrial Economics, Inc. This Population Simulation model is described in Chapter 5 of EPA's "Second Prospective Study – 1990 to 2020 – Benefits and Costs of the Clean Air Act. U.S. Environmental Protection Agency, Office of Air and Radiation, April 2011."

static approach because it explicitly accounts for the year-to-year cascade of impacts on mortality and population following an air pollution change.

To use the PopSim tool, first select **PopSim** from the BenMAP-CE **Tools** menu.

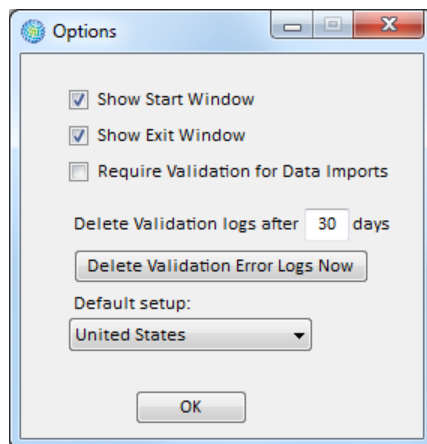


Click through each of the menu items (e.g., **Years**, **Response**, **PM changes**, **Ages affected**, **Lag type**, and **Other**) and review the default settings, or adjust the values as needed. For more information on the underlying datasets or model specifications, please refer to the *“Population Simulation Model for Air Pollution Hazards. Version 3.0. User Manual and Documentation”* (IEC, September 2015).

When you are finished with the model settings, click the **Run model** button on the **Run** screen. A progress bar will be displayed as the calculations are performed. When the model run has completed, a dialog will be displayed. Click **OK** to acknowledge the message. Then click the **Output** button. Specify a location to save the model results. By default, the tool will save to the *“My Documents\My BenMAP-CE Files\PopSim”* folder. The progress bar will once again display as the files are saved to your destination. When complete, a dialog will indicate *“File Saved”*. Click **OK** to acknowledge the message. You may close the PopSim tool, or return to the beginning to initiate a new model run. The tool will remember the settings from the previous run only.

9.11 Options

You can customize some of the generic options for BenMAP-CE using the *Options* window available from the **Tools** menu.



- **Show Start Window:** If unchecked, the **Welcome** window will not appear during subsequent start-ups. You can update your choice at any time.
- **Show Exit Window:** If unchecked, the window confirming you would like to exit will not appear during subsequent shut-downs. You can update your choice at any time.
- **Require Validation for Data Imports:** If checked, you will be required to validate their input files prior to importing datasets. If unchecked, validation will be an available option but it will not be required.
- **Delete Validation logs after __ days:** You can specify the number of days BenMAP-CE will retain the validation logs (the default value is 30 days). If the number of days is left blank, BenMAP-CE will not automatically delete any validation logs.³⁹
- **Delete Validation Error Logs Now:** Select this button if you want to immediately clear all validation logs. *Note: There is no confirmation option here – once the button is clicked, all logs will be deleted.*
- **Default setup:** Select the preferred setup to appear by default in the main BenMAP-CE window.

9.12 Compute Grid Crosswalk

The Compute Grid Crosswalk tool allows users to manually remove crosswalks and re-generate selected crosswalks. A crosswalk is a file used to relate data, such as air quality, population and demographic data, at one spatial scale to another. Crosswalks

³⁹ Validation logs are saved at C:\Users\<User Name>\Documents\My BenMAP-CE Files\ValidationResults.